

Infrastructure as Code with Copilot

DevOps & Infrastructure | Advanced

Agenda

Time	Topic
0:00–0:08	IaC model
0:08–0:20	Terraform module
0:20–0:30	Bicep comparison
0:30–0:42	Module composition
0:42–0:52	Security review
0:52–1:00	Validation and lab handoff

IaC model

Concept	Terraform	Bicep
Resource	<code>resource</code> block	<code>resource</code> declaration
Reusable unit	<code>module</code>	<code>module</code>
Input	<code>variable</code>	<code>param</code>
Output	<code>output</code>	<code>output</code>
Preview	<code>terraform plan</code>	<code>az deployment what-if</code>

Prompt from facts

Read this Terraform project. Add a network module for the supplied CIDRs.
Expose only needed VPC and subnet IDs. Use variables and outputs; do not add
credentials, public database access, or a plan/apply command.

Check every resource against the architecture before accepting it.

Security review

- Block unneeded public access.
- Restrict ingress by port and source.
- Keep secrets out of source.
- Encrypt storage and databases.
- Limit IAM actions and resources.
- Add required logging.

Generated IaC is a draft. Human review owns the deployment decision.

Validate locally

```
terraform fmt
terraform validate
az bicep build --file main.bicep
```

A plan or what-if needs an approved environment, credentials, and stop guard.

Lab

Generate Terraform and Bicep, secure the supplied Terraform, and compare approved Space context with a normal prompt. Do not provision cloud resources.